

AmritChidiya (अमृतचिडिया) — Technical Architecture Handbook

Comprehensive Engineering Specifications, Security Protocols, and Team Matrix (Newbie Friendly)

1. Executive Summary: The Indian Reality (Explained for Beginners)

Every year, the Government of India and various state governments allocate tens of thousands of crores of rupees for student scholarships, farmer income support, women empowerment schemes, and health insurance. Yet, a large percentage of eligible citizens never receive a single rupee. Why?

- Linguistic Exclusion: Most portals use formal English or bureaucratic vocabulary that rural citizens cannot comprehend.
- Opaque Criteria: Eligibility rules depend on multiple variables (annual family income, state domicile, caste category, marks percentage, gender) that confuse citizens.
- The Last-Mile Digital Divide: Rural citizens do not own laptops or printers; they must visit local Jan Seva Kendra / cyber cafe operators who often lack exact scheme details.
- Rampant Cyber Fraud: Fraudulent WhatsApp messages lure vulnerable citizens into paying fake registration fees for welfare schemes that are 100% free.

AmritChidiya (अमृतचिडिया) solves this completely. By serving as an empathetic, multilingual digital companion, it communicates via voice and text in 5 Indian languages, calculates exact financial eligibility, guides citizens step-by-step, warns against cyber scams, and outputs physical printable dossiers for village cyber cafes.

2. The Complete Citizen Journey: Step-by-Step Walkthrough

Here is exactly what happens when a citizen uses AmritChidiya, from start to finish:

Step	Stage	What the Citizen Does & What the System Executes
1	Language Selection	Citizen selects Hindi, Hinglish, English, Marathi, or Tamil. The UI immediately updates all greetings, buttons, and neural voice models.
2	Voice / Text Inquiry	Citizen types or speaks (e.g. 'Main UP ka student hoon, B.Tech 1st year, OBC, income 1.5 lakh'). Audio is transcribed in sub-500ms via Groq Whisper.
3	Yojana Calculator	Citizen can also open the Calculator modal, pick their State, Category, and Income, and instantly see dynamic matched schemes with estimated cash figures (₹30K-₹55K).
4	Intake & Scheme Delivery	LangGraph agent checks for missing details. If complete, it delivers verified scheme cards with official government portal links (e.g. scholarship.up.gov.in).
5	Step-by-Step Guide	Citizen asks how to apply. AmritChidiya provides a structured 5-step registration roadmap with document requirements (Aadhaar, income certificate, bank passbook).
6	Printable CSC Dossier	Citizen clicks CSC Dossier to generate an official printable receipt (AC-CSC-XXXXXX) with an NPCI bank checklist and takes it to their village Jan Seva Kendra.
7	Cyber Scam Verification	If citizen receives a suspicious WhatsApp link (e.g. 'Free Laptop 2026'), they paste it into Cyber Shield. It validates whether it's an authentic .gov.in portal or a scam.
8	Secure Cloud Storage	Chat session is encrypted with Bcrypt and saved to Supabase Cloud PostgreSQL (or local SQLite if offline) for cross-device access.

3. 4 Engineering Pillars & Individual Member USPs

Each team member is the sole technical owner of one critical department:

Member	Role	Individual Standout USP
Member 1	Frontend UI/UX Lead	PhygitaL CSC Dossier Bridge: Generates printable receipts (AC-CSC-XXXXXX) and live Yojana Calculator for village Jan Seva Kendras.
Member 2	Backend API & Speech Lead	Low-Latency Vernacular Audio: High-speed Whisper Large v3 voice pipeline with Indian phonetic bias and neural audio streaming.
Member 3	AI Agent & Cyber Safety Lead	Anti-Hallucination Matcher & Cyber Shield: Zero-hallucination LangGraph engine paired with domain cyber scam inspector.
Member 4	Database & DevOps Lead	Zero-Trust Hybrid DB & Bcrypt Auth: Dual Supabase + SQLite failover layer with salted 12-round Bcrypt security and 24-hr JWT tokens.

4. System Architecture Demystified (Explained with Analogies)

To understand how the technical stack functions under the hood, consider this analogy:

- Frontend (Next.js 16 + React 19) = 'The Citizen Counter': The sleek, interactive touch screen where citizens type, click dropdowns, or view their chat turns.
- Backend API (FastAPI) = 'The High-Speed Post Office': Handles incoming HTTP requests, routes audio files, checks security credentials, and sends back responses in milliseconds.
- AI Core (LangGraph + Groq LLM) = 'The Scheme Expert Brain': Governed by strict state machine rules, it decides what questions to ask and which schemes match.
- Speech Pipeline (Whisper + Edge-TTS) = 'The Ears and Voice': Transcribes Indian vernacular accents with geographic phonetic bias, and speaks back in smooth, natural neural voices.
- Database (Supabase PostgreSQL + SQLite) = 'The Cloud Vault with Offline Backup': Stores encrypted accounts and chat logs in the cloud, while keeping an instant offline replica.

5. Security Architecture (Explained for Beginners)

Government platforms require bank-grade security. Here is how AmritChidiya protects citizen data:

- Bcrypt Password Hashing (12 Rounds): Rather than storing passwords in plain text, passwords are converted into irreversible cryptographic hashes using 12 rounds of salt. Even if an attacker steals the database, they cannot decrypt the passwords.
- Signed JWT Tokens (Digital Wristbands): Upon login, the user receives an HS256-signed JWT token valid for 24 hours. Every private request presents this token, ensuring users can only read or delete their own chats.
- SlowAPI Request Throttling: Protects the server against bots and denial-of-service attacks (5 signups/min, 10 logins/min, 30 chats/min).
- Security Headers & Origin Regex: Enforces strict browser rules ('X-Frame-Options: DENY', 'X-Content-Type-Options: nosniff') and allows secure cross-origin communication between Vercel and Render.

6. Technical Glossary & Key Concepts for Beginners

Term / Acronym	What it Stands For	Beginner-Friendly Explanation
JWT	JSON Web Token	A secure, digital wristband issued upon login that proves who the user is without re-checking their password every time.
Bcrypt	Password Hashing Function	A slow, salted cryptographic algorithm designed specifically to protect passwords from supercomputers and hackers.
STT / TTS	Speech-to-Text / Text-to-Speech	STT turns spoken human voice into computer text. TTS turns computer text into a spoken human voice.
CSC / VLE	Common Service Center / Village Level Entrepreneur	Physical government assistance kiosks in Indian villages where citizens get digital certificates and welfare benefits processed.
NPCI / DBT	National Payments Corporation / Direct Benefit Transfer	The Indian government mechanism where scholarship and welfare cash is deposited directly into a citizen's bank account via Aadhaar.
CORS	Cross-Origin Resource Sharing	A browser security rule that controls whether a website hosted on Vercel is allowed to talk to a backend hosted on Render.

7. Step-by-Step Setup & Cloud Deployment Guide

Local Setup:

1. Clone the repository and navigate into the folder: `git clone https://github.com/ShreyRamola/AmritChidiya.git`
2. Backend: `cd backend && python -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt && python main.py` (Runs on `http://127.0.0.1:8000`)
3. Frontend: `cd frontend && npm install && npm run dev` (Runs on `http://localhost:3000`)

Production Cloud Deployment:

- Frontend (Vercel): Deploy `frontend/` directory with environment variable `NEXT\_PUBLIC\_API\_URL` pointing to the Render backend URL.
- Backend (Render): Deploy `backend/` web service with `ALLOWED\_ORIGINS=https://amrit-chidiya.vercel.app`, `GROQ\_API\_KEY`, `JWT\_SECRET\_KEY`, and Supabase credentials.
- Database (Supabase): Run `supabase\_schema.sql` in Supabase SQL Editor to initialize `users` and `chats` tables with Row Level Security (RLS) policies.